📌 1. Bottom Navigation Bar

ใช้ **BottomNavigationBar** เพื่อสร้างแถบเมนูด้านล่างที่เปลี่ยนหน้าได้

โค้ดตัวอย่าง

dart

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```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatefulWidget {
  @override
  _MyAppState createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  int _selectedIndex = 0;
  static const List<Widget> _widgetOptions = <Widget>[
    Text('Home Page'),
    Text('Profile Page'),
    Text('Settings Page'),
  ];

  void _onItemTapped(int index) {
    setState(() {
      _selectedIndex = index;
    });
```

```
  # use with the CupertinoIcons class f
  cupertino_icons: ^1.0.8
  google_maps_flutter: ^2.10.0
  webview_flutter: ^4.4.4
  geolocator: ^13.0.2
  http: ^0.13.6

dev_dependencies:
  flutter_test:
    sdk: flutter
```

โหลดไปก่อน

เอา api key จาก

https://console.cloud.google.com/

```
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text('Bottom Navigation Example')),
        body: Center(child: _widgetOptions[_selectedIndex]),
        bottomNavigationBar: BottomNavigationBar(
          items: const [
            BottomNavigationBarItem(icon: Icon(Icons.home), label: 'Home'),
            BottomNavigationBarItem(icon: Icon(Icons.person), label: 'Profile'),
            BottomNavigationBarItem(icon: Icon(Icons.settings), label: 'Settings'),
          ],
          currentIndex: _selectedIndex,
          onTap: _onItemTapped,
        ),
      ),
    );
  }
}
```

✅ **Concepts:**

- _selectedIndex ใช้กำหนดหน้าที่เลือก
- onTap ใช้เปลี่ยนหน้า
- BottomNavigationBarItem ใช้สร้างปุ่ม

📌 **2. WebView (ฝังเว็บในแอป)**

ใช้ **flutter_webview_plugin** หรือ **webview_flutter** เพื่อเปิดเว็บในแอป

ติดตั้งแพ็กเกจ

yaml
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```
dependencies:
  webview_flutter: ^4.4.4
```

โค้ดตัวอย่าง

dart
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```
import 'package:flutter/material.dart';
import 'package:webview_flutter/webview_flutter.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: WebViewExample(),
    );
  }
}

class WebViewExample extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("WebView Example")),
```

```
      body: WebView(
        initialUrl: "https://flutter.dev",
        javascriptMode: JavascriptMode.unrestricted,
      ),
    );
  }
}
```

✅ **Concepts:**

- WebView โหลดเว็บจาก initialUrl
- javascriptMode: JavascriptMode.unrestricted ให้เว็บรัน JavaScript ได้

📌 3. Google Map Integration

ต้องใช้ **google_maps_flutter** และตั้งค่า **Google API Key**

ติดตั้งแพ็กเกจ

yaml
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```
dependencies:
  google_maps_flutter: ^2.5.0
```

เพิ่ม API Key (Android)

ไฟล์ AndroidManifest.xml

xml
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```
<manifest>
    <application>
```

📌 **2. กรณีใช้งาน API (Google Maps) บน Android**

ต้องเพิ่ม **Google Maps API Key** ใน `android/app/src/main/AndroidManifest.xml`

xml Copy Edit
```xml
<manifest>
    <application>
        <meta-data
            android:name="com.google.android.geo.API_KEY"
            android:value="YOUR_GOOGLE_MAPS_API_KEY"/>
    </application>
</manifest>
```

```xml
<meta-data android:name="com.google.android.geo.API_KEY"
    android:value="AIzaSyC2VreM81llowMBmoLASCdr7X4n"/>
```

```
        <meta-data
            android:name="com.google.android.geo.API_KEY"
            android:value="YOUR_GOOGLE_MAPS_API_KEY"/>
    </application>
</manifest>
```

โค้ดตัวอย่าง

dart
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```
import 'package:flutter/material.dart';
import 'package:google_maps_flutter/google_maps_flutter.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MapScreen(),
    );
  }
}

class MapScreen extends StatefulWidget {
  @override
  _MapScreenState createState() => _MapScreenState();
}

class _MapScreenState extends State<MapScreen> {
  late GoogleMapController mapController;
```

```
  final LatLng _center = const LatLng(13.736717, 100.523186); // Bangkok

  void _onMapCreated(GoogleMapController controller) {
    mapController = controller;
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Google Map Example")),
      body: GoogleMap(
        onMapCreated: _onMapCreated,
        initialCameraPosition: CameraPosition(target: _center, zoom: 11.0),
      ),
    );
  }
}
```

✅ **Concepts:**

- GoogleMap แสดงแผนที่
- initialCameraPosition กำหนดตำแหน่งเริ่มต้น
- onMapCreated ใช้ควบคุมแผนที่

📌 4. CRUD Database with API

ตัวอย่างใช้ **REST API** (เชื่อมต่อเซิร์ฟเวอร์)
แพ็กเกจที่ใช้: http

โค้ดเทสๆรวมbottom googlemap webview

Main.dart

```
import 'package:flutter/material.dart';
import 'map.dart';
import 'webview.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatefulWidget {
  @override
  _MyAppState createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  int _selectedIndex = 0;

  final List<Widget> _pages = [
    HomeScreen(),    // 🏠 หน้า Home
    MapScreen(),     // 🌍 หน้า Google Maps
    WebViewScreen() // 🌐 หน้า WebView
  ];

  void _onItemTapped(int index) {
    setState(() {
      _selectedIndex = index;
    });
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(title: Text('Flutter Final Exam')),
        body: _pages[_selectedIndex],
        bottomNavigationBar: BottomNavigationBar(
          items: const [
            BottomNavigationBarItem(icon: Icon(Icons.home), label: 'Home'),
            BottomNavigationBarItem(icon: Icon(Icons.map), label: 'Map'),
            BottomNavigationBarItem(icon: Icon(Icons.web), label: 'Web'),
          ],
          currentIndex: _selectedIndex,
          onTap: _onItemTapped,
        ),
      ),
    );
  }
}

// ✅ **หน้า Home**
class HomeScreen extends StatelessWidget {
  @override
```

```
  Widget build(BuildContext context) {
    return Center(
      child: Text("🏠 Welcome to Home Screen!", style: TextStyle(fontSize:
20)),
    );
  }
}
```

Map.dart ฉบับยังไม่ได้ระบุพิกัด ฉบับเบื้องต้น

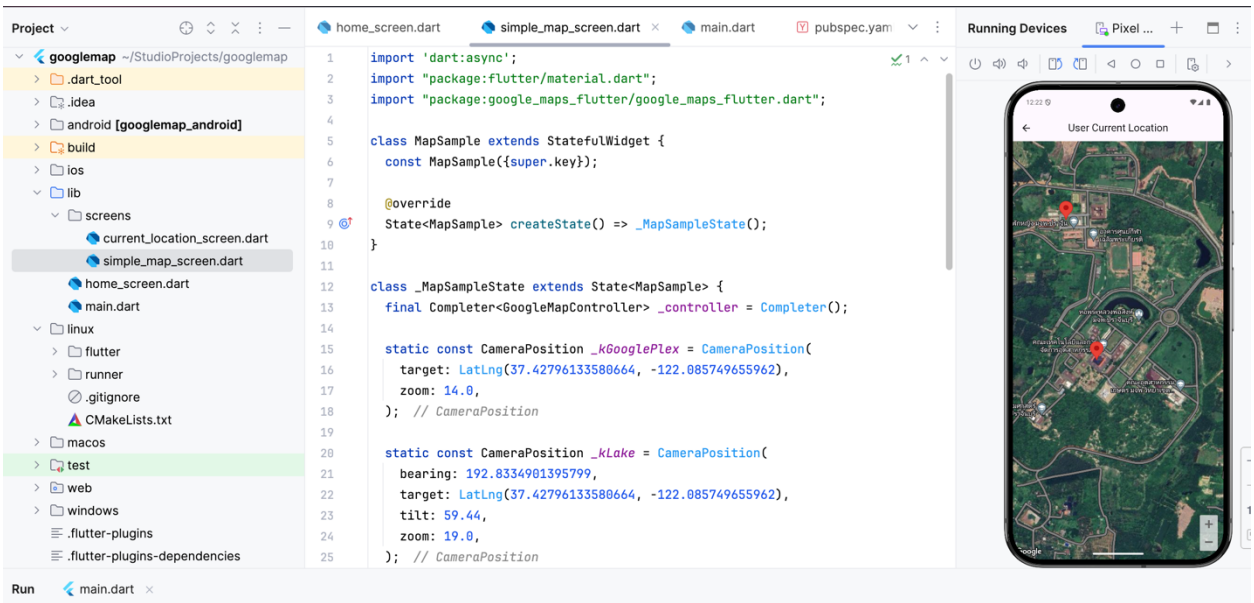
```
import 'package:flutter/material.dart';
import 'package:google_maps_flutter/google_maps_flutter.dart';

class MapScreen extends StatefulWidget {
  @override
  _MapScreenState createState() => _MapScreenState();
}

class _MapScreenState extends State<MapScreen> {
  late GoogleMapController mapController;
  final LatLng _initialPosition = LatLng(13.736717, 100.523186); // พิกัดกรุงเทพฯ

  void _onMapCreated(GoogleMapController controller) {
    mapController = controller;
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: GoogleMap(
        onMapCreated: _onMapCreated,
        initialCameraPosition: CameraPosition(target: _initialPosition, zoom:
11.0),
      ),
    );
  }
}
```

Webview.dart

```
import 'package:flutter/material.dart';
import 'package:webview_flutter/webview_flutter.dart';

class WebViewScreen extends StatefulWidget {
  @override
  _WebViewScreenState createState() => _WebViewScreenState();
}

class _WebViewScreenState extends State<WebViewScreen> {
  late final WebViewController _controller;

  @override
  void initState() {
```

```
    super.initState();
    _controller = WebViewController()
      ..loadRequest(Uri.parse('https://flutter.dev'))
      ..setJavaScriptMode(JavaScriptMode.unrestricted);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: WebViewWidget(controller: _controller),
    );
  }
}
```

Google Map FUllllll



Main.dart

```
import 'dart:async';
import "package:flutter/material.dart";
import "package:google_maps_flutter/google_maps_flutter.dart";
import "package:googlemap/home_screen.dart";

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomeScreen(),
    );
  }
}
```

home_screen.dart

```
import 'package:flutter/cupertino.dart';
import 'package:flutter/material.dart';
import 'package:googlemap/main.dart';
import 'package:googlemap/screens/simple_map_screen.dart';
import 'package:googlemap/screens/current_location_screen.dart';

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});

  @override
  State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Google Map New "),
        centerTitle: true,
      ),
      body: SizedBox(
        width: MediaQuery.of(context).size.width,
        child: Column(
          children: [
            ElevatedButton(
                onPressed: () {
                  Navigator.of(context)
                      .push(MaterialPageRoute(builder: (BuildContext context)
{
                    return MapSample();
                  }));
                },
                child: const Text("Simple Map")),
            ElevatedButton(
                onPressed: () {
                  Navigator.of(context)
                      .push(MaterialPageRoute(builder: (BuildContext context)
{
                    return CurrentLocation();
                  }));
                },
                child: const Text("Current Location"))
          ],
        ),
      ),
    );
  }
}
```

Folder Screen

Simple_map_screen.dart

```
import 'dart:async';
import "package:flutter/material.dart";
import "package:google_maps_flutter/google_maps_flutter.dart";

class MapSample extends StatefulWidget {
  const MapSample({super.key});

  @override
  State<MapSample> createState() => _MapSampleState();
}

class _MapSampleState extends State<MapSample> {
  final Completer<GoogleMapController> _controller = Completer();

  static const CameraPosition _kGooglePlex = CameraPosition(
    target: LatLng(37.42796133580664, -122.085749655962),
    zoom: 14.0,
  );

  static const CameraPosition _kLake = CameraPosition(
    bearing: 192.8334901395799,
    target: LatLng(37.42796133580664, -122.085749655962),
    tilt: 59.44,
    zoom: 19.0,
  );

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Google Maps"),
        leading: IconButton(
          icon: Icon(Icons.arrow_back),
          onPressed: () => Navigator.pop(context), // BACK ปุ่มย้อนกลับ
        ),
      ),
      body: GoogleMap(
        mapType: MapType.hybrid,
        initialCameraPosition: _kGooglePlex,
        onMapCreated: (GoogleMapController controller) {
          _controller.complete(controller);
        },
      ),
      floatingActionButton: FloatingActionButton.extended(
        onPressed: _goToTheLake,
        label: const Text("Go to the Lake"),
        icon: const Icon(Icons.directions_boat),
      ),
    );
  }

  Future<void> _goToTheLake() async {
    final GoogleMapController controller = await _controller.future;
    controller.animateCamera(CameraUpdate.newCameraPosition(_kLake));
  }
}
```

current_location_screen.dart

```
import 'package:flutter/material.dart';
import 'package:geolocator/geolocator.dart';
import "package:google_maps_flutter/google_maps_flutter.dart";

class CurrentLocation extends StatefulWidget {
  const CurrentLocation({super.key});

  @override
  State<CurrentLocation> createState() => _CurrentLocationState();
}

class _CurrentLocationState extends State<CurrentLocation> {
  late GoogleMapController googleMapController;

  static const CameraPosition _initialCameraPosition =
  CameraPosition(target: LatLng(14.1593588, 101.3463206), zoom: 16);

  Set<Marker> markers = {};
  final LatLng _posi1 = const LatLng(14.1651265, 101.34394);
  final LatLng _posi2 = const LatLng(14.1589163, 101.3453382);
  final LatLng _posi3 = const LatLng(14.1650378, 101.3377752);

  void _onMapCreated(GoogleMapController controller) {
    googleMapController = controller;
    _addMarker(_posi1, 'Dormitory boy');
    _addMarker(_posi2, 'Digital');
    _addMarker(_posi3, 'Airport');
  }

  void _addMarker(LatLng position, String markerId) {
    setState(() {
      markers.add(Marker(
          markerId: MarkerId(markerId),
          position: position,
          infoWindow: InfoWindow(title: markerId)));
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("User Current Location"),
        centerTitle: true,
      ),
      body: GoogleMap(

        markers: markers,
        mapType: MapType.hybrid,
        initialCameraPosition: _initialCameraPosition,
        // onMapCreated: (GoogleMapController controller) {
        //   googleMapController = controller;

        // },
        onMapCreated: _onMapCreated,
      ),
```

```
    );
  }

  Future<Position> _determinePosition() async {
    bool serviceEnabled;
    LocationPermission permission;

    serviceEnabled = await Geolocator.isLocationServiceEnabled();
    if (!serviceEnabled) {
      return Future.error("Location services are disabled.");
    }

    permission = await Geolocator.checkPermission();
    if (permission == LocationPermission.denied) {
      permission = await Geolocator.requestPermission();
      if (permission == LocationPermission.denied) {
        return Future.error("Location permission denied.");
      }
    }

    if (permission == LocationPermission.deniedForever) {
      return Future.error("Location permission permanently denied.");
    }

    return await Geolocator.getCurrentPosition();
  }
}
```